

Earth

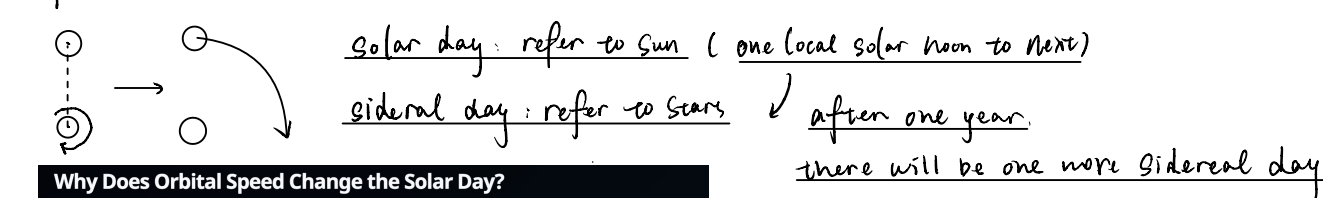
Why study Earth in Astronomy?

- 1. It is easy to study.
- 2. It's where we live
- 3. Earth & other planets shares some fundamental mechanism of formation serves as a baseline
- 4. We need to know how earth move before observing stars
Helps us understand the stars

A Day: 24 Hours

Definition: Earth rotates one cycle (360°)

reference: Sun/Moon/star

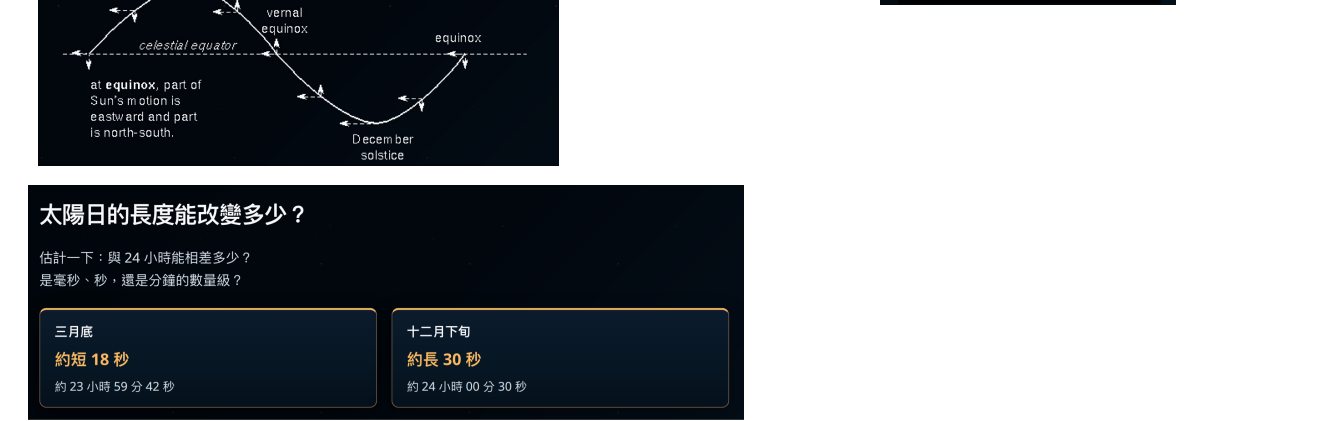
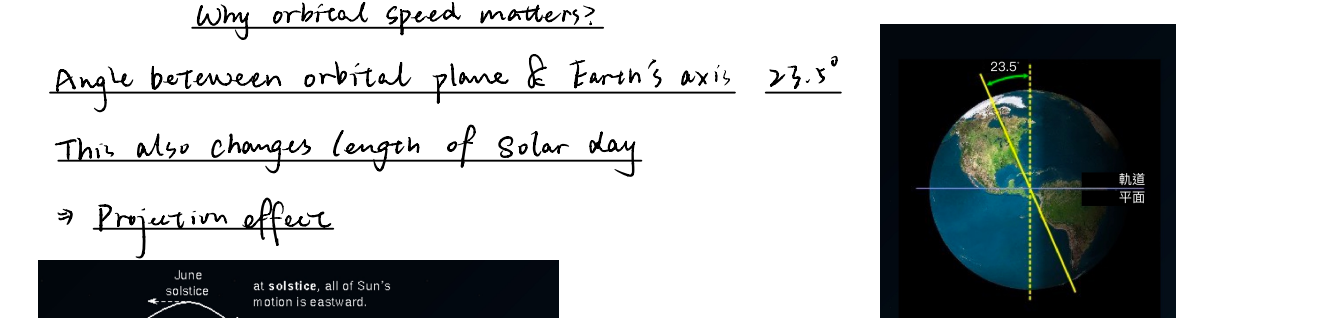


Why Does Orbital Speed Change the Solar Day?

Diagram illustrating the relationship between Earth's rotation and its orbital motion around the Sun. It shows how the combination of these two motions results in the 24-hour solar day.

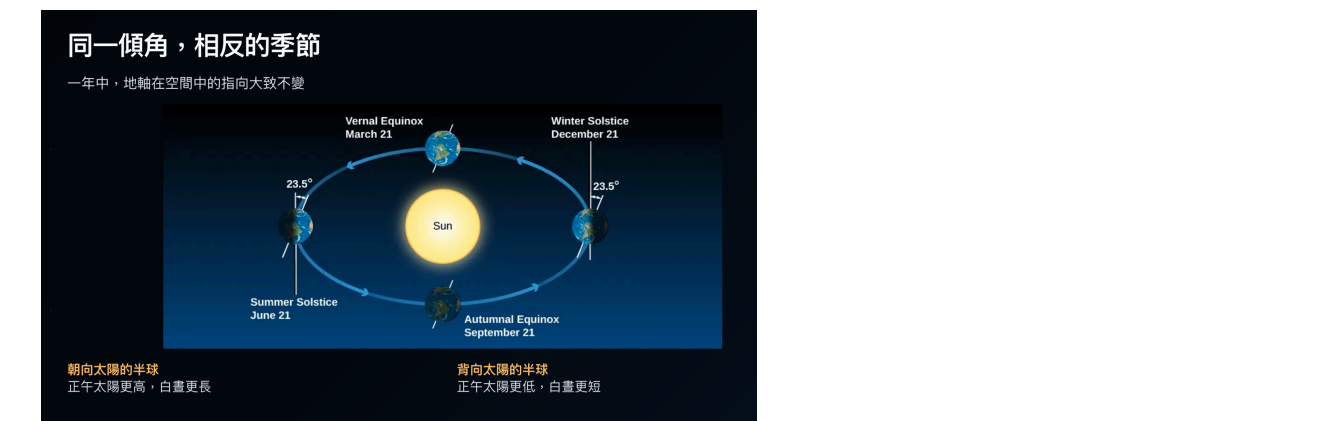
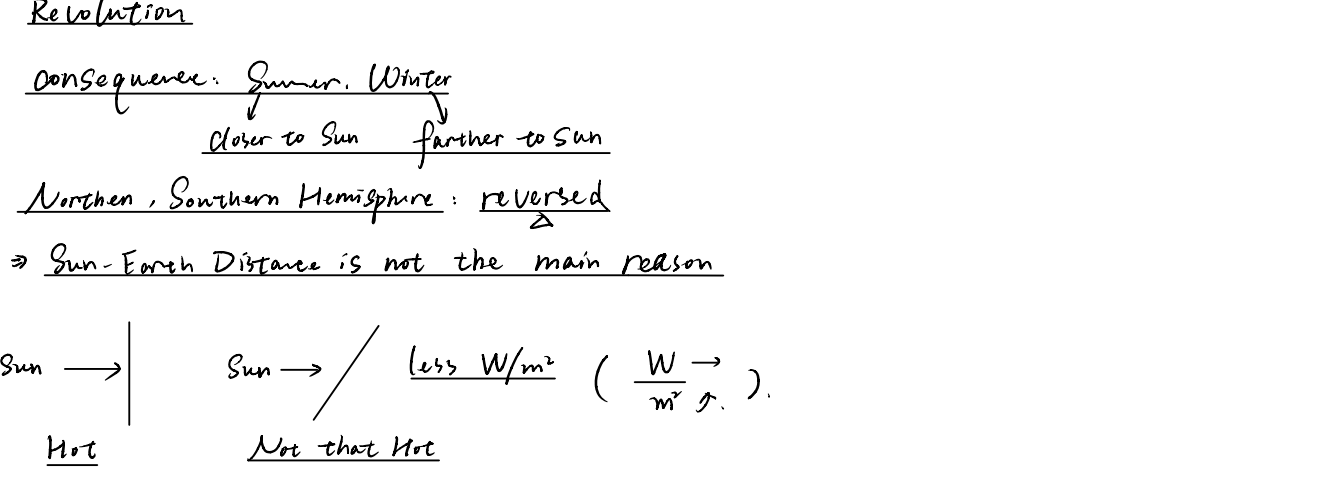
1. 一个太阳日 = 24 小时 (平均而言, 忽略闰秒带来的“一天”)
2. 一个恒星日 = 23 小时 56 分 4 秒。
3. 为什么会有这个差别?
关键在于: 地球在自转的同时, 还在绕太阳公转。
• 地球自转一圈 (相对于恒星) 后, 已经绕了约 1° 的轨道弧。
• 为了再次“对准”太阳, 地球必须再转约 1°, 这需要约 4 分 36 秒。
• 所以, 太阳日 = 恒星日 + 约 3 分 56 秒。

mean solar day
why solar day varies?
→ Different orbital speed at different time
Why orbital speed matters?
Angle between orbital plane & Earth's axis: 23.5°
This also changes length of solar day
→ Projection effects



How Earth Moves
revolution: earth rotating around the Sun
→ Summer/Winter...
rotation: rotation of earth
→ solar day / sidereal day
As South/North Star don't move

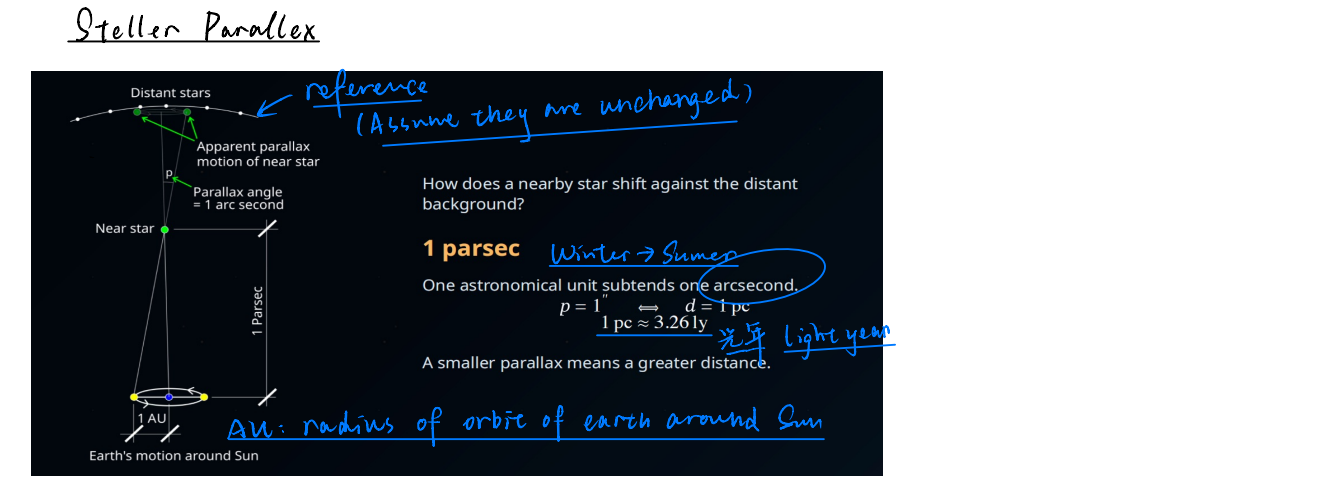
Revolution
consequence: Summer, Winter
closer to Sun, further to Sun
Northern, Southern Hemisphere: reversed
→ Sun-Earth Distance is not the main reason



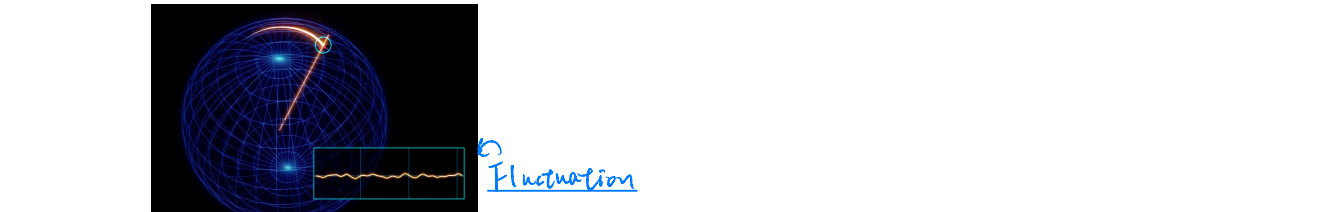
Distance: $\frac{147^2}{15^2} = 0.84$
Angle: $\cos 23.5 = 0.9$
Since we must consider latitude!
Summer: 16.5° Big difference
Winter: 63.5°

Southern Hemisphere Has more water
Cooler in Summer than northern

Stellar Parallax



Axial Precession
 $L = F \times r$ Angular Momentum Conservation
precession of a spinning top



Constellation 昴星
algebra 彗星



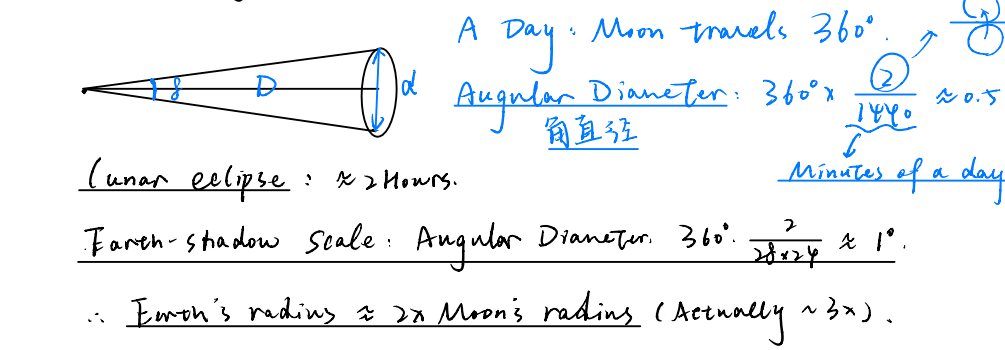
Zodiac: which constellation lies behind the sun?
Why Babylonians label is wrong? precession changes the sky

Moon

- 1. Distance 384400 km. (1s for light travel)
- 2. Radius 1737.4 km.
- 3. Mass 7.35×10^{22} kg. Density: 60% of Earth
- 4. Formation ≈ 4.5 Ga

How Early We can estimate Moon's radius

- 1. Read the Curved Shadow
→ Moon is much smaller than shadow of the Earth (Equal to radius of Earth)
The ratio of Earth's shadow & Moon
→ We know Radius of Earth → Estimate Radius of Moon.
Moon's Angular Diameter

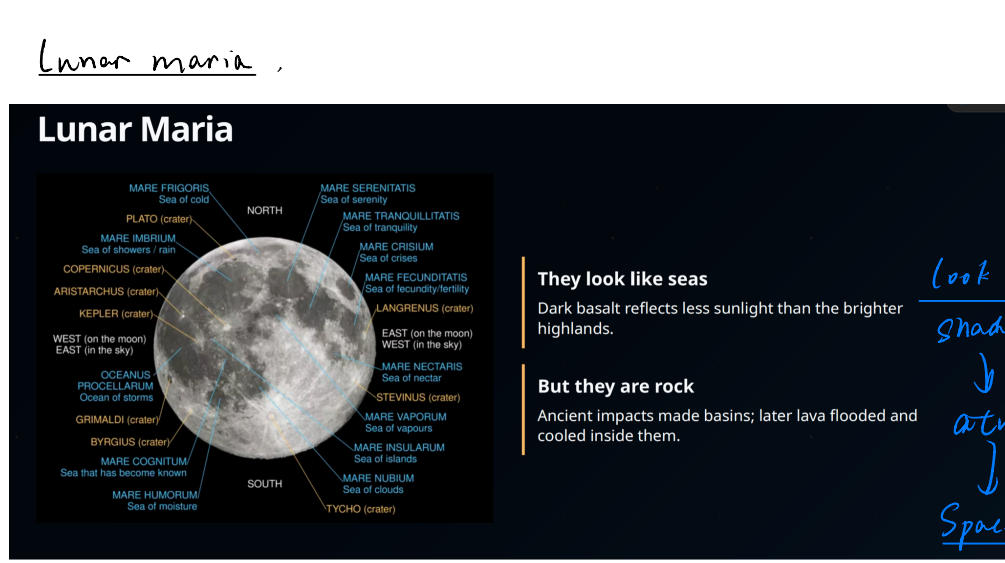


Moon's Mass
Size + Density → Mass.
($\approx 1/3$ Earth) (assume: rock)

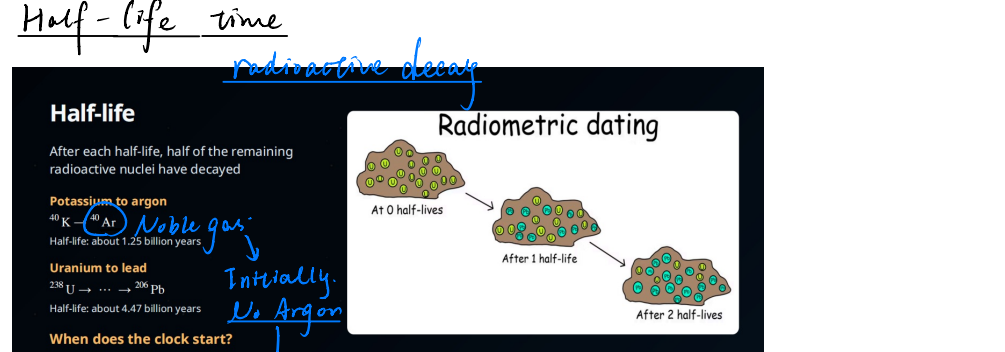
Newton's gravity, Tide 3:2: compare Sun/Moon
in order of magnitude of side
Moon = $M_0 \left(\frac{d_{moon}}{d_0} \right)^3 \approx 10^{22}$ kg
Mass of sun distance.

How Can we know more precisely
Distance: using a reflection on moon
Mass: using a satellite orbits moon
→ period, orbitalize → gravitational parameters
→ Measure surface gravity & radius
→ Mass

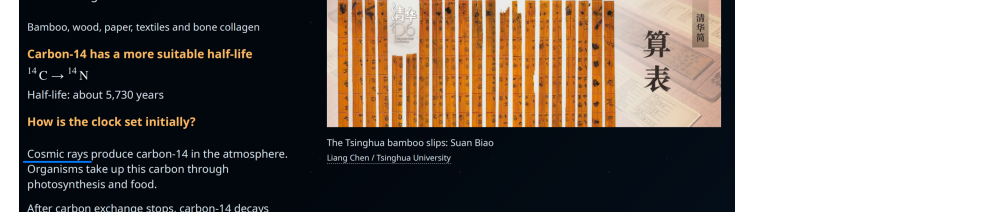
Lunar maria



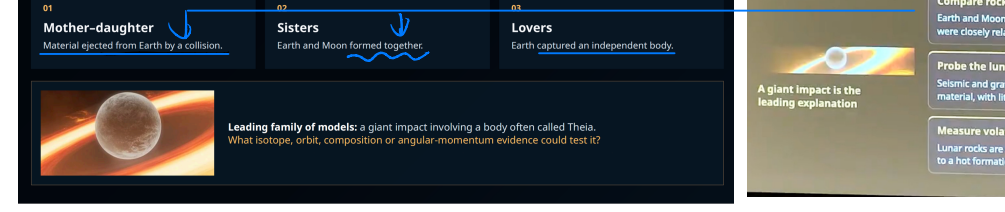
Moon's age 4.5 Billion year
get moon's rock
Half-life time



Archaeology
Carbon-14 has a more variable half-life
We measure how many argon ions
→ The age of the rock



What's the Origin of the Moon?
Mother daughter
Sisters
Lovers



Does the Moon Rotate Around Earth?
Close-up: Earth
The whole orbit



Why two Faces of moon differ
At Beginning Earth/Moon both hot.
Tide Locking: The face toward Earth: Hotter
rock → liquid → smoother.
The back Side: More craters (further)

Moon's Familiar face
coincidence? No. many other planet also has tide locking

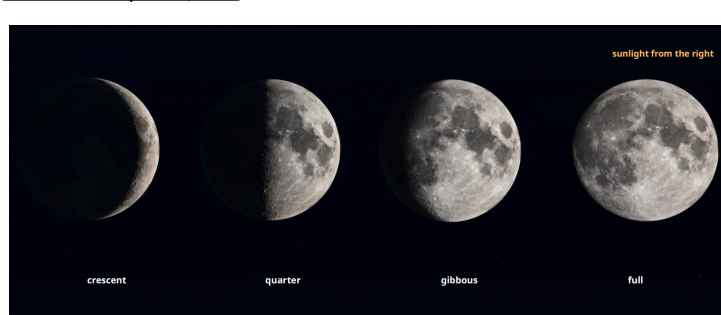
Tidal Locking
A small deformation
The gravitational torque with a delay, Earth
pulls on the moon's bulge, slowing a rapidly rotating moon.
Energy dissipates as heat
Repeated losing energy. Over time, the spin settles into one face to the other.
Stretch → Heat → loss energy



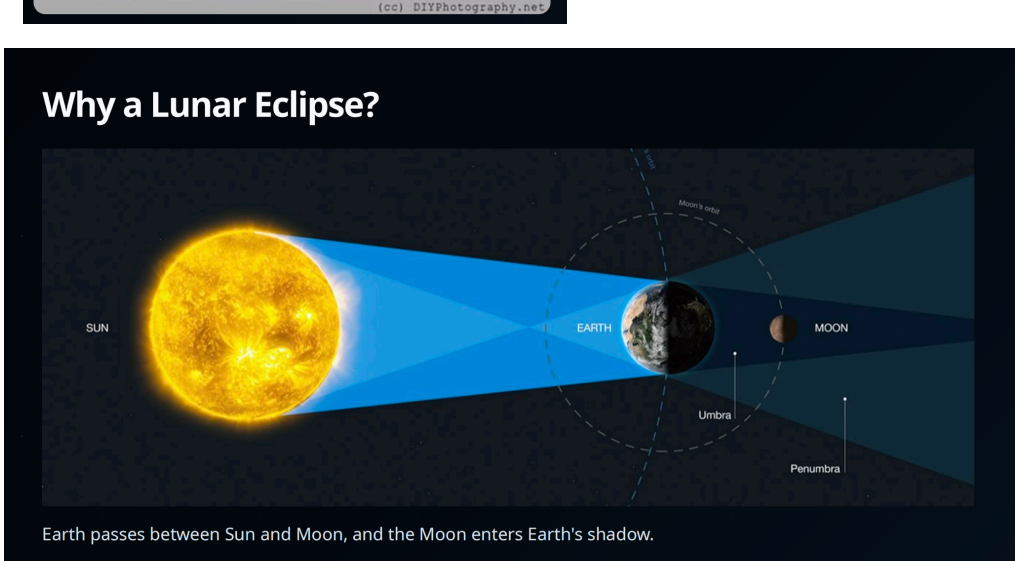
Difference of sky from Earth & Moon
1. No twinkling stars: No atmosphere
2. More stars. Sun light don't screen
3. Day & Night: Each lasts ≈ 2 weeks
Colder than Earth

Moon-Earth-Sun Relation

Moon Phases



Difference with eclipse: Earth blocked the light (~ 2 hours)
Moon phase: ~ 1 month
Sun shining at different angle



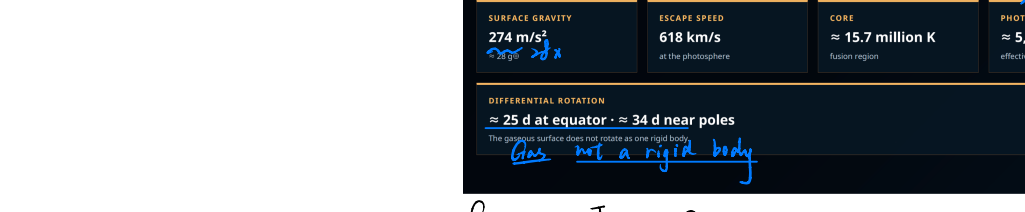
Why a Lunar Eclipse?
Earth passes between Sun and Moon, and the Moon enters Earth's shadow.

Sun
The Sun — Scale
The Sun — Conditions



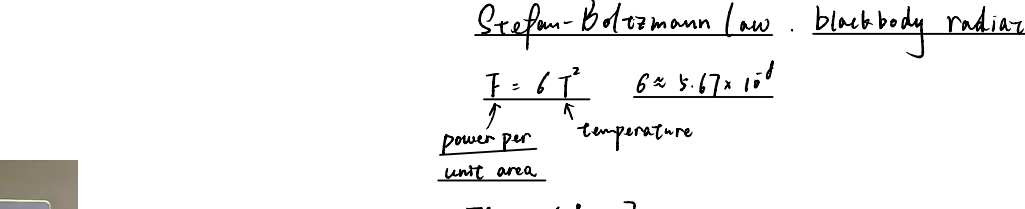
Shining: Energy Output
→ Wind, hydro food, fossil fuels. Solar...
from the Sun
Total Power of Sun
Above atmosphere at 1 AU
 $F_0 = 1361 \text{ W/m}^2$
 $L_0 = 4\pi (1 \text{ AU})^2 F_0 \approx 3.83 \times 10^{26} \text{ W}$
Stefan-Boltzmann law: blackbody radiation
 $F = \sigma T^4$
power per unit area
 $T(0) = 6000 \text{ K}$
Infering Solar Power from Temperature
What can we infer without touching the object?
Solar luminosity
Surface area
Power per unit area
 $L_0 = 3.83 \times 10^{26} \text{ W}$
Stars: main sequence, planets, the early universe

Where does Solar power come from?
Historical ideas: Chemicals (fire too short),
Matter impacts,
Gravitational contraction
...



What Does the Sun Actually Burn?
The Sun is a ball of gas, mostly hydrogen and helium. The core is where the nuclear fusion takes place, converting hydrogen into helium and releasing energy.

Sun-Earth Distance: 1 AU.
How to measure?



Which Stars Should We Measure First?
Need distance
Need a target
Break the loop with imperfect stars.
Start with bright and high proper motion stars, then measure parallax.

